

AYAAN SANADI

Innovative Machine Learning and Computer Vision Engineer specializing in deploying high-performance deep learning models, Edge AI, and real-time kinematic tracking systems. Proven track record of optimizing neural networks for resource-constrained hardware and architecting end-to-end intelligent pipelines that drive actionable physical-world analytics.

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CORE TECHNICAL SKILLS

Python, JavaScript, C, C++, YOLO, MediaPipe, TensorFlow Lite (Micro), PyTorch, FastAPI, OpenCV, Next.js, Git, PostgreSQL, MongoDB

PROFESSIONAL EXPERIENCE

Computer Vision Intern

Jan 2026 – Apr 2026

Neuronetic Vision

- Engineered end-to-end object detection pipelines by scraping, cleaning, and annotating complex image datasets of **[Insert Number]** images, improving training data quality and reducing model training convergence time by **[Insert %]**.
- Trained, fine-tuned, and deployed YOLO models for real-time CCTV security systems, achieving a detection accuracy of **[Insert %]** mAP for weapons (knives, handguns) and public spitting violations under varied lighting conditions.
- Developed automated vision solutions for warehouse and facility management, deploying models that achieved a **[Insert %]** recall rate in identifying pests and detecting two-wheeler license plates to streamline security auditing.

KEY ENGINEERING PROJECTS

FormCheck AI: Real-Time Biomechanics Analysis

- Architected an AI-driven gym safety platform leveraging MediaPipe and OpenCV to perform real-time kinematic tracking and pose estimation for compound lifts, processing video feeds at **[Insert FPS]** FPS.
- Delivered instant corrective audio cues with a latency of under **[Insert Milliseconds]** ms by implementing a custom validation pipeline that cross-referenced user form against domain-specific biomechanical metrics.

BirdSpec: End-to-End Species Detection Pipeline

- Classified 483 bird species with **[Insert %]** accuracy by training a lightweight EfficientNet-V2 CNN on a dataset of 60,000+ images, successfully resolving high inter-class similarity and morphological differences.
- Containerized and deployed the inference engine via a FastAPI service, integrating a Next.js front-end and Supabase backend to reduce query response times to **[Insert Milliseconds]** ms for user history tracking.

Real-Time AI Smart Attendance & Analytics Pipeline

- Automated classroom attendance for 70+ students by engineering a production-ready computer vision pipeline utilizing YOLOv8-face and InceptionResnetV1 for live facial feature extraction, achieving **[Insert %]** recognition accuracy.
- Maintained a high processing speed of **[Insert FPS]** FPS and eliminated ID-switching by architecting a 'Two-Brain' tracking system pairing short-term physics tracking (DeepOCSORT/Kalman filters) with long-term logical memory.
- Accelerated real-time video processing by **[Insert %]** by offloading tensor operations to Apple Silicon (MPS backend) and matching live frames to a vectorized master database via cosine similarity.
- Streamlined administrative reporting by integrating a Supabase (PostgreSQL) backend, reducing manual registry logging time by **[Insert %]** through automated AI output translation and dynamic CSV generation.

AI-Powered Acoustic Pest Detection System (Edge AI / TinyML)

- Detected agricultural pest infestations in real-time by deploying an INT8 post-training quantized CNN with a footprint of under 65KB onto an ESP32 microcontroller using TFLite Micro, reducing power consumption by ****[Insert %]****.
- Replicated Python's Librosa library at the edge by building a custom C++ Digital Signal Processing (DSP) pipeline, converting raw 16kHz audio into Mel-Spectrograms on-device in under ****[Insert Milliseconds]**** ms.
- Prevented micro-controller RAM overflow and hardware-level tensor corruption by manually forcing 16-byte memory alignment and dynamically shifting massive audio arrays to the Heap, achieving ****[Insert %]**** system uptime.
- Eliminated false-positive alerts by ****[Insert %]**** through a custom 5-step rolling buffer with '3-out-of-5' voting logic, pushing verified HTTP POST alerts to a responsive Next.js dashboard.

EDUCATION

BMS Institute of Technology and Management

2024 – 2028

B.E. in Artificial Intelligence & Machine Learning | CGPA: 8.6